Relation Extraction with Graph Neural Networks
Thesis Phase 1 for B.Tech Project

Kesa Varshitha (CB.EN.U4CSE19121)
N D N Jahnavi (CB.EN.U4CSE19135)
R Sneha Nair (CB.EN.U4CSE19139)
Sithara J P (CB.EN.U4CSE19148)

Project Mentor: Bindu K.R., Asst. Prof
Dept. of Computer Science & Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham

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Abstract

Entity-relationship extraction extracts semantic relationships from a text. These train the NLP-based AI models to understand the relationships with multiple entities in a text for the analysis of the extracted data. Since graph neural network (GNN) can well learn the relationship between data elements in the document, it is proposed to use GNN to solve the problem of entity recognition (NER) and relation- ship extraction in documents. The relation extraction task performs poorly on complex text. To solve this problem, we propose a novel joint model named extracting Entity-Relations via Improved Graph Attention networks, which enhances the ability of the relation extraction task. Our model would demonstrate consistent performance improvements, without requiring heavy feature engineering nor additional language-specific knowledge.

Keywords

Relation extraction, NER, GNN

1 Acknowledgments

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2 Introduction

An organization may implement AI-driven decision-making processes to enhance their efficiency by incorporating heterogeneous documents. To automate these transactions successfully, we must be able to incorporate semantic understanding, in addition to the traditional Optical Character Recognitions (OCR). Information extraction (IE) is the process of automatically retrieving structured data from semi-structured and scanned machine-readable documents. This requires named entity recognition (NER) and the discovery of relationships between document terms. NER identifies word or phrase spans in unstructured text (the entity) and classify them as belonging to a particular class (the entity type). Relation Extraction (RE) identifies relationships implied in the text between a pair of entities. NER and relation extraction (RE) are two important tasks in information retrieval.

We propose a method to extract structured information from semi structured documents using GNNs. Among existing models, graph neural networks (GNNs) is one of the most effective approaches for multi-hop relational reasoning.

2.1 Motivation

There is a considerable quantity of unstructured electronic material available on the Web. A common concept is to arrange unstructured text by appending semantic information. Due to large data, human annotation is not practicable. Instead, we would want a computer to automatically annotate every data with the structure we are interested in. This could be easily done using relation extraction. Relationship extraction is the task of extracting semantic relationships from a text. Due to incorrect extraction of data, an NLP algorithm could be trained incorrectly, therefore extracting the most relevant relations in the texts is important. The majority of NLP models are done mainly on three languages. Our main goal is to use an improved version of this methodology to translate Indian languages.

2.2 Major Contribution

- The project aims at developing a novel method for relation extraction from semi-structured documents by means of GNNs.
- GP-GNN enables relational message-passing with rich text information, which could be applied to process relational reasoning on unstructured inputs.
- The proposed model, GP-GNN, solves the relational message-passing task by encoding natural language as parameters and performing propagation from layer to layer.
- The proposed model is compared both qualitatively and quantitatively against state-of-the-art relation extraction models in the literature.
- The project casts the named entity recognition and relation extraction as a supervised message passing task.
- The project surpass state-of-the-art performance of the three tasks involved. These tasks are word grouping, entity labelling and entity linking.
- The model generalizes to weakly structured documents.

3 Literature Survey

[1] Zhu, H., Lin, Y., Liu, Z., Fu, J., Chua, T.S. and Sun, M., 2019. Graph neural networks with generated parameters for relation extraction. arXiv preprint arXiv:1902.00756.
Algorithm - Graph neural network with generated parameters (GPGNNs)

[2] Cetoli, A., Bragaglia, S., O'Harney, A.D. and Sloan, M., 2017. Graph convolutional networks for named entity recognition. arXiv preprint arXiv:1709.10053.
Algorithm - Bidirectional LSTM, Bi - Graph Convolutional Networks (Bi-GCN)

[3] Chiu, J.P. and Nichols, E., 2016. Named entity recognition with bidirectional LSTM-CNNs. Transactions of the association for computational linguistics, 4, pp.357-370.
Algorithm - Bidirectional LSTM, Character-level CNN

[4] Carbonell, M., Riba, P., Villegas, M., Fornés, A. and Lladós, J., 2021, January. Named entity recognition and relation extraction with graph neural networks in semi structured documents. In 2020 25th International Conference on Pattern Recognition (ICPR) (pp. 9622-9627)
Algorithm - k-NN graph, BERT

[5] Sorokin, D. and Gurevych, I., 2017, September. Context-aware representations for knowledge base relation extraction. In Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing (pp. 1784-1789).
Algorithm - LSTM-based encoder

[6] Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C. and Philip, S.Y., 2020. A comprehensive survey on graph neural networks. IEEE transactions on neural networks and learning systems, 32(1), pp.4-24.
Algorithm - Recurrent graph neural networks, Convolutional graph neural networks

[7] Dess`ı, D., Osborne, F., Recupero, D.R., Buscaldi, D. and Motta, E., 2021. Generating knowledge graphs by employing natural language processing and machine learning techniques within the scholarly domain. Future Generation Computer Systems, 116, pp.253-264.
Algorithm - CSO TOOL, Word2vec algorithm

[8] Zeng, W., Lin, Y., Liu, Z. and Sun, M., 2016. Incorporating relation paths in neural relation extrac-tion. arXiv preprint arXiv:1609.07479.
Algorithm - Text Encoder, Relation Path Encoder, Joint Model

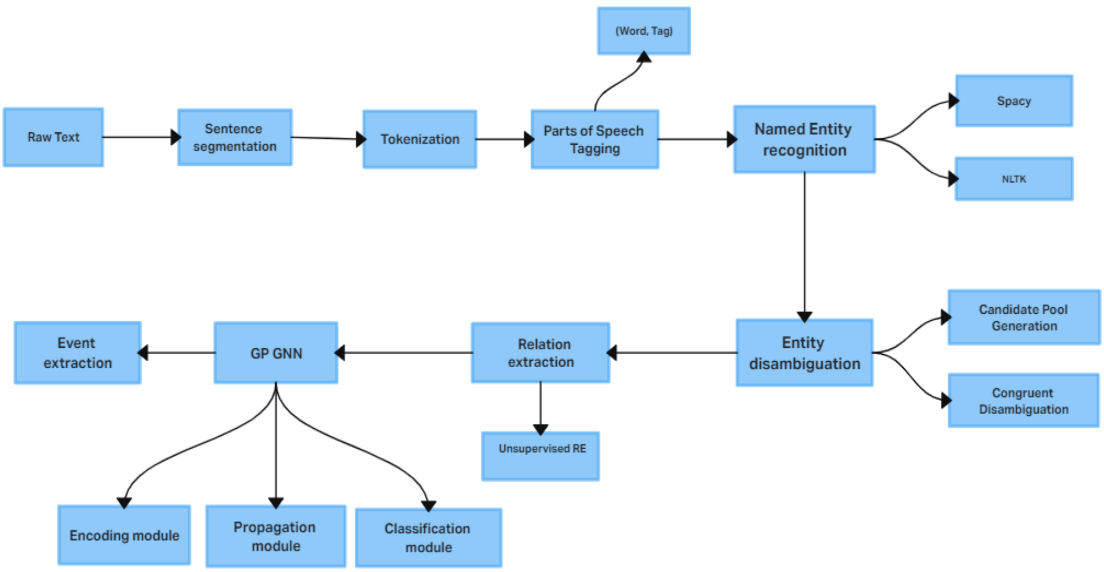
[9] Harrando, I. and Troncy, R., 2021, June. Named entity recognition as graph classification. In Euro-pean Semantic Web Conference (pp. 103-108). Springer, Cham.
Algorithm - Binary Auto-encoder, TransE, Node2Vec, GCN

3.1 Inferences From the Literature

- Methods like RNN understand text as a one dimensional. Collection of input vectors; any syntactic information namely the parse tree of the sentence is ignored.
- BERT, the computational resources needed to train/fine-tune and make inferences. And BERT will promote the results which is in dictionary or in Wikipedia it cannot be trained in a way like other models.
- LSTMs take longer to train. LSTMs require more memory to train. Does not make any difference between the abbreviation and the word.
- CNNs are specially built to operate on regular (Euclidean) structured data.
- Most existing GNNs can only process multi-hop relational reasoning on pre-defined graphs and cannot be directly applied in natural language relational reasoning.
- We conclude that graph neural networks with generated parameters (GP-GNNs) is one of the most effective approaches for multi-hop relational reasoning in relation extraction, enabling GNNs to process relational reasoning on unstructured text inputs.

4 Proposed System

- **Architecture Diagram**



- **Modules**

1. Dataset preparation and preprocessing
2. Tokenization
3. Pos Tagging
4. Word grouping (keyword detection): Detecting document entities, i.e., grouping words with the same semantics
5. Handling Unknown pos tagging
6. Entity classification: Divides the detected entities into preset categories
7. Relationship prediction: Find pairing relationship between entities.
8. GP GNN algorithm

- **Algorithm - GP GNN**

Graph neural networks with generated parameters (GP-GNNs) adapt graph neural networks to solve the natural language relational reasoning task. GP-GNNs first constructs a fully connected graph with the entities in the sequence of text. After that, it employs three modules to process relational reasoning: (1) an encoding module (2) a propagation module (3) a classification module. As compared to traditional GNNs, GP-GNNs could learn edge parameters from natural languages, extending it from performing inference on only non-relational graphs or graphs with a limited number of edge types to unstructured inputs such as texts.

A graph neural network with generated parameters (GPGNNs) is one of the most effective approaches for multi-hop relational reasoning in relation extraction.

5 Results and Analysis

Sentence segmentation
The sentences are segmented into words when a punctuation is encountered.

Tokenization
The paragraphs and sentences are split into smaller units (words) that can be more easily assigned meaning. The keywords were generated. The stop words in Hindi were dropped and the punctuations were removed.

Parts of Speech Tagging (PoS)
PoS tagging converts a word to a list of tuple (word, tag). Since it was a Hindi dataset, there were unknown tags. It was handled using google translate.

Named Entity Recognition (NER)
BIO Tagging
Each token is tagged to one of the seven standard NLP BIO – Beginner, Inside, Outside - tags (I - Location, B - Organization, I - Person, O, B - Person, I - Organization, B – Location). Using the BIO tags, numeric feature is extracted. It is then converted to numeric feature vectors.

Bi - LSTM for Named Entity Recognition (NER)
Accuracy and validation accuracy are increasing as the epoch number increases. This shows that the model is working well. Loss and validation loss are decreasing from 1.7627 to 1.2389.

Decision Tree
Decision tree classifies the tokens to the 7 tags and checks the accuracy of the model. The F1 score accuracy is 90%.

6 Conclusion and Future Work

We conclude that graph neural networks with generated parameters (GP-GNNs) is one of the most effective approaches for multi-hop relational reasoning in relation extraction which enables GNNs to process relational reasoning on unstructured text inputs.

Our proposed model, GP-GNN, solves the relational message-passing task by encoding natural language as parameters and performing propagation from layer to layer. GP GNN can be considered as a more generic framework for graph generation problem with unstructured input other than text. We demonstrated its effectiveness in predicting the relationship between entities in naturallanguage.

Future work includes method and testing on larger data sets which could confirm the feasibility of the approach as a generic solution for extracting structured information from semi-structured documents.

7 References

[1] Zhu, H., Lin, Y., Liu, Z., Fu, J., Chua, T.S. and Sun, M., 2019. Graph neural networks with generated parameters for relation extraction. arXiv preprint arXiv:1902.00756.

[2] Cetoli, A., Bragaglia, S., O'Harney, A.D. and Sloan, M., 2017. Graph convolutional networks for named entity recognition. arXiv preprint arXiv:1709.10053.

[3] Chiu, J.P. and Nichols, E., 2016. Named entity recognition with bidirectional LSTM-CNNs. Transactions of the association for computational linguistics, 4, pp.357-370.

[4] Carbonell, M., Riba, P., Villegas, M., Fornés, A. and Lladós, J., 2021, January. Named entity recognition and relation extraction with graph neural networks in semi structured documents. In 2020 25th International Conference on Pattern Recognition (ICPR) (pp. 9622-9627)